

EXPONENTIAL HITTING-TIME ASYMPTOTICS FOR EIGENVALUE GAPS IN DYSON–ORNSTEIN–UHLENBECK DYNAMICS, VIA BESSEL COUPLING

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ABSTRACT. For the Dyson–Ornstein–Uhlenbeck process at general $\beta > 1$, we prove that a fixed bulk spectral gap's survival probability past time T is exponentially asymptotic, $P(I_i(T) = 1) = e^{-(1+o(1))Kq} + o(1)$, with an explicit rate $q \sim \kappa_\beta(n\varepsilon)^{\beta+1}$ derived from cubic-type level repulsion via a Girsanov coupling to a Bessel process of dimension $\beta+1$, valid for $\varepsilon = n^{-C}$ in an explicit range $C \in (1, C^*(\beta))$, $C^*(\beta) = (2\beta+3)/(2\beta+2)$, that narrows as β grows. The normalization $n\varepsilon$ — rather than the $\sqrt{n\varepsilon}$ one might naively guess by analogy with CLT-scale fluctuations — is forced by elementary counting against the equilibrium measure's semicircle support, confirmed independently by direct simulation (batched Euler–Maruyama on GUE matrices, $n = 100, 400, 1600$). An auxiliary estimate needed along the way — bounding the probability of hitting the threshold having started a block already close to it — reduces, once correctly placed in the small-time (rather than large-time) asymptotic regime of the underlying Bessel hitting formula, to a stretched-exponentially small correction via a direct SDE comparison argument. The same local counting argument extends to non-semicircle bulk densities: we give the analogous marginal statement for a fixed pair of eigenvalues of a Wishart–Ornstein–Uhlenbeck sample covariance process, with an application to the loss of local resolution between two specific financial risk factors (§5.4).

A natural strengthening of the marginal statement above is a *cover time* law: the last of the n bulk gaps to first go below ε , standardized, converging to a Gumbel distribution — the dynamical analogue of a coupon-collector phenomenon. We discuss this as an open direction in §5. The natural strategy (coupling each gap to an idealized independent Bessel proxy and transferring the resulting Poisson approximation to the true joint count) provably fails, in four independent ways that all reduce to the same divergent quantity — ruling out that specific strategy rather than merely falling short of it. A more promising route (local dependence à la Chen–Stein, applied directly to the true joint process) reduces the remaining difficulty to gaps at exactly unit spatial separation (an exact computation shows the driving noises of any two gaps two or more apart are already independent), for which we identify an explicit, uniformly positive drift lower bound on the product of adjacent gaps (via an AM–GM inequality applied to their exact Itô coupling), and, for $\beta \geq 2$, prove that the *reciprocal* of that product is an exact nonnegative supermartingale, giving via Ville's maximal inequality a complete, closed-form, numerically verified bound on the probability that two adjacent gaps are simultaneously near their threshold at any time. This resolves, for $\beta \geq 2$, the specific comparison-theorem target left open in earlier work on this note; it is a global (rather than small-time) estimate and, by a direct order-of-magnitude comparison, is likely not yet sharp enough on its own to complete the joint theorem, and $1 < \beta < 2$ remains untouched by this argument.

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1. INTRODUCTION

1.1. Motivation: from random matrices to dynamical extremes. Random matrix theory began with Wigner’s observation that the statistical fluctuations of nuclear energy levels match those of the eigenvalues of large random matrices. In the seventy years since, random matrix universality has been discovered in an extraordinary range of settings: zeros of the Riemann zeta function, spectra of quantum-chaotic billiards, lengths of increasing subsequences of random permutations, and correlation functions in two-dimensional Yang–Mills theory [18, 14]. The unifying feature is *level repulsion*: eigenvalues behave like charged particles repelling each other, with repulsion exponent $\beta \in \{1, 2, 4\}$ for the classical ensembles (real, complex, quaternion).

Despite this richness, the existing theory is almost entirely *static*: it describes the eigenvalue distribution at a fixed time or a fixed matrix. The dynamical question — how do eigenvalues move when the matrix is perturbed stochastically over time? — is far less understood. The natural model is the *Dyson Brownian motion* (Dyson–Ornstein–Uhlenbeck, or Dyson–OU), a system of n interacting diffusions

$$(1) \quad d\lambda_i = -\frac{1}{2}\lambda_i dt + \frac{1}{\sqrt{n}} dB_i + \frac{1}{n} \sum_{j \neq i} \frac{dt}{\lambda_i - \lambda_j}, \quad i = 1, \dots, n,$$

introduced by Dyson in 1962 [8] and studied extensively since. The Coulomb repulsion term $(\lambda_i - \lambda_j)^{-1}$ enforces level repulsion: the eigenvalues cannot cross, and the nearest-neighbour gaps $g_i = \lambda_{i+1} - \lambda_i$ are pushed away from zero with strength proportional to β . At equilibrium, the gaps obey $P(g_i \leq s) \asymp \kappa_G s^3$ for small s at $\beta = 2$ (the famous cubic level repulsion; see [14], Ch. 7).

The question motivating this paper is: *in the Dyson–OU dynamics, how long does it take for every bulk spectral gap to have experienced a near-collision?* This note establishes

the underlying marginal statistics — how long a *single* fixed gap takes to first near-collide — as a complete, general- β theorem (§5), and discusses the joint (all-gaps-at-once) strengthening that would answer the motivating question in full as an open direction (§5.3).

This is a *cover time* question, entirely analogous to asking how long a random walk takes to visit every site of a graph. For random walks, the answer is well known: the cover time of an n -vertex graph scales as $n \log n$ and its fluctuations follow a Gumbel law (Aldous [3], Ding–Lee–Peres [6]). Here we ask the same question for the eigenvalue dynamics: fix a small threshold $\varepsilon > 0$ and call the moment $g_i(t) \leq \varepsilon$ a *near-collision* of gap i . The *eigenvalue cover time* is

$$(2) \quad \tau_{\text{cov}}^\varepsilon := \max_{i \in I_{\text{bulk}}} \inf\{t \geq 0 : g_i(t) \leq \varepsilon\},$$

the last time any bulk gap experiences its first near-collision.

1.2. The main result, and an open strengthening. The main result of this note (Theorem 5.3, §5) is a marginal hitting-time asymptotic: for a fixed bulk gap i and any $\beta > 1$, the probability that gap i never near-collides before time T is $\exp(-(1 + o(1))Kq_\beta) + o(1)$, with an explicit rate $q_\beta \sim \kappa_\beta(n\varepsilon)^{\beta+1}$ and an explicit admissible range for ε . The proof combines a Girsanov coupling of the gap process to a Bessel process of dimension $\beta + 1$ with the Abadi–Galves rare-event theorem, after a dimensional-analysis correction (§3.2) to the normalisation of gap universality: elementary counting against the equilibrium measure forces n -scaled, not $\sqrt{n}$ -scaled, gaps, a point confirmed independently by direct simulation on GUE matrices. §5.4 gives a second application of the same theorem to a non-semicircle bulk density (Wishart–OU sample covariance dynamics), with a specific financial reading in terms of risk-factor pairs losing local resolution.

An open strengthening: the eigenvalue cover time. A natural and considerably stronger statement is a joint one across all $\asymp n$ bulk gaps simultaneously: the standardised cover time (2),

$$r(\varepsilon, n) \tau_{\text{cov}}^\varepsilon - \log n \Rightarrow \text{Gumbel},$$

the dynamical analogue of the coupon-collector phenomenon underlying classical cover-time theorems for random walks (Aldous [3], Ding–Lee–Peres [6]). This would follow from Theorem 5.3 together with a joint Poisson approximation for the survival indicators $\{I_i(T)\}_{i \in I_{\text{bulk}}}$ ($I_i(T) = 1$ iff gap i has not near-collided by time T); §5.3 identifies precisely where the natural strategy for such a joint statement currently stops, narrows the obstruction to a specific two-gap comparison problem, and records partial progress on it (an explicit drift bound, a matching comparison process, and a numerical check ruling out one candidate complication).

1.3. Relation to the existing literature. Static random matrix extremes. Ben Arous–Bourgade [4] (minimum gap Weibull), Forrester [14] (gap distributions, β -ensembles), Erdős–Yau [11] (gap universality).

Dynamical random matrices. Dyson [8] (1962), Erdős–Schlein–Yau [10] (rigidity), Bourgade–Erdős–Yau [5] (circular β -ensembles).

Cover times and extreme values. Aldous [3], Ding–Lee–Peres [6], Dembo–Peres–Rosen–Zeitouni [7] (Brownian motion); Leadbetter–Lindgren–Rootzén [17] (stationary processes); Abadi–Galves [1] (mixing processes).

Bessel processes and RMT. Revuz–Yor [19], Karatzas–Shreve [16] (BES); König–Eichelsbacher [15] (conditioned walks).

1.4. Organisation. Section 2 collects general, scale-independent facts about Bessel processes and states the level-repulsion hypothesis (H_β). Section 3 introduces the Dyson-OU process, its equilibrium measure, and the external RMT inputs, including the correctly normalised gap-universality statement (§3.2). Section 4 gives the high-probability control machinery (uniform-in-time rigidity, far-field cancellation) underlying the main proof. Section 5 derives the rate formula and proves the main theorem (Steps 1–2, §5.4 for the Wishart-OU application), then turns to the open joint extension (Step 3, §5.3): the state of the joint-transfer obstruction, the specific two-gap comparison problem it reduces to, and the partial progress recorded on that problem.

2. BACKGROUND: BESSEL PROCESSES AND LEVEL REPULSION

2.1. Bessel processes. Several key arguments in the intended proof strategy (§5) — a Girsanov coupling reducing the gap SDE to a Bessel process, and a hitting-probability computation for the one-block rate — reduce the behaviour of a spectral gap to that of a Bessel process. The connection between Bessel processes, level repulsion, and wall-wetting phenomena goes back to Fisher [13]. We collect the relevant definitions and facts from Revuz–Yor [19] (Chapter XI) and Karatzas–Shreve [16] (Section 5.5).

Definition 2.1 (Bessel process of dimension d). Let $d \geq 1$ and $x_0 > 0$. A *Bessel process of dimension d* started at x_0 is the unique strong solution of

$$(3) \quad dX_s = \frac{d-1}{2X_s} ds + dW_s, \quad X_0 = x_0 > 0,$$

where W is a standard Brownian motion. Equivalently, if d is a positive integer, $X_s = \|B_s\|$ where B is d -dimensional Brownian motion started at a point of norm x_0 .

Remark 2.2 (Non-explosion and polar behaviour for $d > 2$). For $d \geq 2$ the SDE (3) has a unique positive strong solution ([19], XI.1.5; [16], 5.5.C). For $d > 2$ the origin is polar: the process is immediately repelled by the drift $(d-1)/(2x)$, and for any fixed $x_0 > 0$ and $r < x_0$,

$$P\left(\inf_{0 \leq s \leq \delta} X_s \leq r \mid X_0 = x_0\right) \leq C\left(\frac{r}{x_0}\right)^{d-2} \quad ([19], \text{XI.1.11}; [16], 5.5.C).$$

This is the basis for the no-clustering argument in the intended proof strategy (§5).

Proposition 2.3 (Scaling and hitting of small levels). *Let X be $\text{BES}(d)$ with $d = \beta + 1 \geq 2$, diffusion $\sigma = 2/\sqrt{\beta}$, started at $x_0 > 0$.*

- (i) (Brownian scaling.) *For $\delta > 0$, the rescaled process $Y_u := X_{u\delta}/\sqrt{\delta}$ is again $\text{BES}(d)$ started at $x_0/\sqrt{\delta}$. In particular, $P(\inf_{s \leq \delta} X_s \leq \varepsilon)$ depends on x_0 and ε only through the ratios $x_0/\sqrt{\delta}$ and $\varepsilon/\sqrt{\delta}$; this is the Brownian scaling property used to compare excursions at different levels.*
- (ii) (Hitting probability from above; [19], XI.1; [16], 5.5.) *If $x_0 \geq c$ (bounded below) and $\varepsilon \ll x_0$, then for $d > 2$ (i.e., $\beta > 1$, not necessarily integer) and any $\delta > 0$:*

$$(4) \quad P\left(\inf_{0 \leq s \leq \delta} X_s \leq \varepsilon \mid X_0 = x_0\right) \asymp \left(\frac{\varepsilon}{x_0}\right)^{d-2} \wedge 1 = \left(\frac{\varepsilon}{x_0}\right)^{\beta-1} \wedge 1,$$

where the implied constants depend on δ/ε^2 (for large δ , the hitting probability is essentially the transience asymptotics and becomes $\asymp (\varepsilon/x_0)^{\beta-1}$; for small $\delta \ll x_0^2$ the probability is negligible). (An earlier application of this bound to the one-block hit probability, with $x_0 = G_0 \asymp n^{1/2}$, used the scaling flagged as wrong in §3.2; redoing it with the corrected $G_0 = O(1)$ is part of §5, Step 1.) For $d = 2$ (i.e., $\beta = 1$), $\text{BES}(2)$ is recurrent and the hitting probability is 1.

- (iii) (Repulsion after a near-hit; [16], 5.5.C.) *For the rescaled process G_i (satisfying an SDE of the form (3)), if $G_i(s_0) = \mathcal{E} = n\varepsilon$ and the block length in local time is $\mathbf{b}_{\text{loc}} = n^2\varepsilon^2$, then by Brownian scaling of $\text{BES}(\beta + 1)$ (rescaling $x \mapsto x/\mathcal{E}$ and $s \mapsto s/\mathcal{E}^2$ reduces to a fixed $\text{BES}(\beta + 1)$ over a fixed time interval):*

$$(5) \quad P(G_i(s_0 + \mathbf{b}_{\text{loc}}) \geq c_\beta \mathcal{E}) \geq 1 - p_0,$$

for universal constants $c_\beta > 1$ and $p_0 \in (0, 1)$ depending only on β . The Bessel repulsion forces the rescaled gap away from $\mathcal{E}$ within one block.

2.2. A key hypothesis: level repulsion.

Definition 2.4 (Hypothesis (H_β)). We say that (H_β) holds if there exist constants $\kappa_\beta > 0$, $\eta > 0$, and $C > 0$ independent of n and $i \in I_{\text{bulk}}$ such that for all sufficiently small ε in the rare near-collision regime (to be made precise in §5),

$$(6) \quad P(g_i(0) \leq \varepsilon) = \kappa_\beta (\sqrt{n}\varepsilon)^{\beta+1} + O((\sqrt{n}\varepsilon)^{\beta+1+\eta})$$

uniformly in $i \in I_{\text{bulk}}$.

Remark 2.5 (Explicit remainder rates). For $\beta = 2$, Theorem 3.4 implies that one can take $\eta = c$ for some absolute constant $c > 0$ (the rate of convergence in gap universality from [11]). For $\beta = 1, 4$, the Pfaffian structure of the circular ensembles [14] yields $\eta = 2$ (i.e., the next term in the small- s expansion of the gap distribution is $O(s^{\beta+3})$). For general $\beta > 0$, Bourgade–Erdős–Yau universality [5] (for the circular β -ensembles) is expected to establish (H_β) with a rate depending on the BEY convergence exponent; the explicit value $\eta = 2$ for non-classical β is expected from the tridiagonal model but is not yet proved.

With the Bessel background and the level-repulsion hypothesis in place, we turn to the Dyson-OU process itself and recall the external random matrix theory inputs that underpin the entire argument.

3. THE DYSON–ORNSTEIN–UHLENBECK PROCESS AND EXTERNAL RMT INPUTS

3.1. The Dyson-OU process and spectral gaps. Let $n \geq 2$ and $\beta \geq 1$. The Dyson-OU process $(\lambda_i(t))_{i=1}^n$ is the unique strong solution of

$$(7) \quad d\lambda_i(t) = \sqrt{\frac{2}{\beta n}} dB_i(t) + \frac{1}{n} \sum_{j \neq i} \frac{dt}{\lambda_i(t) - \lambda_j(t)} - \frac{1}{2} \lambda_i(t) dt, \quad i = 1, \dots, n,$$

with $\lambda_1(t) < \dots < \lambda_n(t)$ always ordered. Its unique invariant measure is the Hermite β -ensemble

$$d\mu_{n,\beta}(\lambda) \propto \exp\left(-\frac{\beta n}{4} \sum_{i=1}^n \lambda_i^2\right) \prod_{i < j} |\lambda_i - \lambda_j|^\beta d\lambda.$$

Throughout we assume $\lambda(0) \sim \mu_{n,\beta}$.

The i -th spectral gap is $g_i(t) := \lambda_{i+1}(t) - \lambda_i(t) > 0$. Subtracting the SDE for λ_i from that of λ_{i+1} gives the gap SDE

$$(8) \quad dg_i = \sqrt{\frac{2}{\beta n}} (dB_{i+1} - dB_i) + \frac{2}{ng_i} dt - \frac{1}{n} \sum_{j \neq i, i+1} \frac{g_i dt}{(\lambda_{i+1} - \lambda_j)(\lambda_i - \lambda_j)} - \frac{1}{2} g_i dt.$$

The sign and the factor $1/n$ in the third term are derived directly from equation (7): subtracting the SDE for λ_i from that of λ_{i+1} yields the identity $1/(\lambda_{i+1} - \lambda_j) - 1/(\lambda_i - \lambda_j) = -g_i/[(\lambda_{i+1} - \lambda_j)(\lambda_i - \lambda_j)]$ (negative, so the non-nearest eigenvalues compress the gap), with the overall coefficient $1/n$ from equation (7). The classical location γ_i is the i/n -quantile of the semicircle density $\varrho_{\text{sc}}(x) := \frac{1}{2\pi} \sqrt{4 - x^2} \mathbf{1}_{|x| \leq 2}$. The *rigidity exponent*

$\alpha_{\text{rig}} > 0$ is the universal constant from Erdős–Schlein–Yau [10] and Erdős–Knowles–Yau–Yin [9]; on the bulk rigidity event $\Omega_{\text{rig}}(t)$ one has

$$(9) \quad |\lambda_i(t) - \gamma_i| \leq \frac{C(\log n)^{\alpha_{\text{rig}}}}{n^{1/2}} \quad \forall i \in I_{\text{bulk}}.$$

The *locality scale* is $L := (\log n)^{\alpha_{\text{rig}}+1}$; all far-field Coulomb interactions beyond distance L will be shown to cancel in bulk (Lemma 4.6).

3.2. Microscopic rescaling and local time. The Dyson-OU gap SDE (8) has diffusion coefficient $2/\sqrt{\beta n}$ (after combining the two independent increments $dB_{i+1} - dB_i = \sqrt{2} d\tilde{W}$) and leading Bessel drift $2/(ng_i)$. Both coefficients decay as $n \rightarrow \infty$, reflecting that gaps are microscopic. To work with $O(1)$ coefficients in the intended proof strategy (§5) we introduce the *rescaled gap* $G_i(s) := ng_i(s/n)$ and *local time* $s := nt$:

$$(10) \quad G_i(s) := ng_i(s/n), \quad s := nt.$$

A direct substitution in (8), using $d\tilde{W}_{s/n} = (1/\sqrt{n}) dV_s$ where V is a standard Brownian motion in local time s , yields

$$(11) \quad dG_i = \frac{2}{\sqrt{\beta}} dV_s + \frac{2}{G_i} ds + \tilde{R}_i(s) ds - \frac{G_i}{2n} ds,$$

where $\tilde{R}_i(s)$ encodes the near- and far-field Coulomb interactions with non-neighbouring eigenvalues. On $\Omega_{\text{RMT}}(T)$, rigidity gives $|\tilde{R}_i| \leq C/n$ uniformly (the far-field contribution is $O(1/n)$ in local time by Lemma 4.6, and the near-field similarly). The OU term $-G_i/(2n) = O(1/n)$ is also negligible.

The intended proof strategy (§5) works in local time s with the rescaled gap G_i , with a *rescaled threshold* $\mathcal{E} := n\varepsilon$ and a *block length* $\mathbf{b}_{\text{loc}}$ (local-time analogue of some block length $\mathbf{b}$ in original time) still to be fixed once the range of ε is corrected.

A consistency check worth making explicit. It is tempting to read the gap universality statement as " $\sqrt{n}g_i \rightarrow \text{Gaudin}$," reasoning by analogy with CLT-scale ($\sqrt{n}$) fluctuations elsewhere in random matrix theory — which would give $g_i \asymp n^{-1/2}$, hence $G_i = ng_i \asymp n^{1/2}$. This directly contradicts the derivation two paragraphs above: the SDE for G_i has $O(1)$ coefficients (in particular the drift $2/G_i$) only if G_i itself is typically $O(1)$, which forces $g_i \asymp n^{-1}$, not $n^{-1/2}$ — as elementary counting against the $O(1)$ -support semicircle equilibrium measure confirms directly (Theorem 3.1). With the correctly normalised gap universality ($ng_i \rightarrow \text{Gaudin}$, Theorem 3.4), $G_i = ng_i$ is indeed $O(1)$ typically, consistent with the dimensional analysis above. Every threshold comparison below ($\mathcal{E} \ll G_i$ or otherwise) is made against this $O(1)$ typical value of G_i .

3.3. External results from random matrix theory.

Theorem 3.1 (Bulk rigidity, [10] Theorem 3.1; [9] Theorem 2.2). *Let $(\lambda_1(t) \leq \dots \leq \lambda_n(t))$ be the Dyson β -ensemble started in equilibrium and let $\gamma_1 < \dots < \gamma_n$ be the classical locations. There exist $\alpha_{\text{rig}} > 0$ and $C_0 < \infty$ such that for all $D > 0$ and large n ,*

$$P\left(\exists i \in I_{\text{bulk}}, \exists t \in [0, T] : |\lambda_i(t) - \gamma_i| > \frac{C_0(\log n)^{\alpha_{\text{rig}}}}{\sqrt{n}}\right) \leq n^{-D}.$$

Theorem 3.2 (Log-Sobolev inequality, [12]). *The Hermite β -ensemble $\mu_{n,\beta}$ satisfies a log-Sobolev inequality: there exists $\kappa_{\text{LSI}} > 0$, independent of n , such that for all smooth f ,*

- (i) $\text{Ent}_\mu(f^2) \leq \frac{2}{\kappa_{\text{LSI}}} \mathbb{E}_\mu\left[\frac{1}{\beta n} \sum_i (\partial_i f)^2\right];$
- (ii) $\text{Var}_\mu(f) \leq \frac{1}{\kappa_{\text{LSI}}} \mathbb{E}_\mu\left[\frac{1}{\beta n} \sum_i (\partial_i f)^2\right];$
- (iii) $\text{Var}(e^{tL} f) \leq e^{-2\kappa_{\text{LSI}} t} \text{Var}(f).$

Theorem 3.3 (Local semicircle law, [9]). *For an $n \times n$ Wigner matrix H and resolvent $G(z) = (H - z)^{-1}$ at $z = E + i\eta$ with $|E| \leq 2 - \kappa$ and $\eta \geq n^{-1+\xi}$, $|m_n(z) - m_{\text{sc}}(z)| \leq C_\xi/(n\eta)$ with probability $1 - n^{-D}$.*

Theorem 3.4 (Gap universality, [11]). *For any bulk $i \in I_{\text{bulk}}$ and continuous bounded $f : \mathbb{R}^k \rightarrow \mathbb{R}$,*

$$|\mathbb{E}_{\mu_{n,2}}[f(n g_i, \dots, n g_{i+k-1})] - \mathbb{E}_{\text{sine}}[f(s_0, \dots, s_{k-1})]| \leq C_f/n^c$$

uniformly in $i \in I_{\text{bulk}}$, for some $c > 0$.

Remark 3.5 (On the normalisation). The n -scaling above (rather than $\sqrt{n}$, which appears in some unrelated CLT-type normalisations elsewhere in random matrix theory) is forced already by elementary counting: the equilibrium measure $\mu_{n,\beta}$ has semicircle support $[-2, 2]$ with n eigenvalues, so typical spacing is $g_i \asymp 1/(n\rho_{\text{sc}}(\gamma_i)) = O(1/n)$; the local semicircle law (Theorem 3.3) is likewise stated at the $\eta \geq n^{-1+\xi} \approx 1/n$ microscale. It is also the unique normalisation consistent with §3.2's rescaling $G_i := n g_i$, which requires G_i itself to be $O(1)$ for the gap SDE's coefficients to be $O(1)$. All three checks agree, and are confirmed directly by simulation (batched Euler–Maruyama on GUE matrices, $n = 100, 400, 1600$: $n g_i$ has empirical mean stable at ≈ 2.2 – 2.3 across a $16\times$ range in n , while $\sqrt{n} g_i$ shrinks by half each time n quadruples).

Theorems 3.1–3.4 are the only external inputs from random matrix theory; everything else in the paper is self-contained. The next step is to combine them into a single high-probability event on which all estimates hold uniformly in time.

4. HIGH-PROBABILITY RMT GOOD EVENTS

4.1. Uniform rigidity on the time interval.

Lemma 4.1 (Uniform rigidity on $[0, T]$). *Under the assumptions of Theorem 3.1, for any $D > 0$ and $T \leq e^{n^c}$ with $c > 0$ sufficiently small,*

$$P\left(\sup_{t \in [0, T]} \max_{i \in I_{\text{bulk}}} |\lambda_i(t) - \gamma_i| > \frac{C_0(\log n)^{\alpha_{\text{rig}}}}{\sqrt{n}}\right) \leq n^{-D}.$$

Proof. We discretize the time interval $[0, T]$ into $M = n^A$ points $t_j = jT/M$ with A to be chosen. By Theorem 3.1, for each fixed t_j ,

$$P\left(\max_{i \in I_{\text{bulk}}} |\lambda_i(t_j) - \gamma_i| > \frac{C_0(\log n)^{\alpha_{\text{rig}}}}{\sqrt{n}}\right) \leq n^{-D-2}.$$

A union bound over the M grid points gives

$$P\left(\exists j : \max_{i \in I_{\text{bulk}}} |\lambda_i(t_j) - \gamma_i| > \frac{C_0(\log n)^{\alpha_{\text{rig}}}}{\sqrt{n}}\right) \leq M n^{-D-2}.$$

Now we control the fluctuations between grid points. By Lemma 4.4, for any $i \in I_{\text{bulk}}$ and any t, s with $|t - s| \leq T/M$,

$$P(|\lambda_i(t) - \lambda_i(s)| > n^{-B}) \leq C \exp(-cn^{2B}M/T).$$

Choosing A large enough so that $M = n^A$ and B such that $n^{2B}M/T \geq n$, the right-hand side is $O(e^{-cn})$, which is $o(n^{-D-2})$. A union bound over i and over intervals between grid points yields

$$P\left(\sup_{|t-s| \leq T/M} \max_i |\lambda_i(t) - \lambda_i(s)| > n^{-B}\right) \leq n \cdot M \cdot O(e^{-cn}) \leq n^{-D-2}.$$

Combining the two estimates, and noting that $n^{-B} \ll (\log n)^{\alpha_{\text{rig}}}/\sqrt{n}$, we obtain the desired uniform bound. $\square$

4.2. Definition of the good event.

Definition 4.2 (Good event $\Omega_{\text{RMT}}(T)$). Let $L := (\log n)^{\alpha_{\text{rig}}+1}$. Define

$$(12) \quad \Omega_{\text{RMT}}(T) := \left\{ \sup_{t \in [0, T]} \max_{i \in I_{\text{bulk}}} |\lambda_i(t) - \gamma_i| \leq \frac{C_0 (\log n)^{\alpha_{\text{rig}}}}{\sqrt{n}} \right\} \cap \Omega_{\text{reg}}(T),$$

where

$$\Omega_{\text{reg}}(T) := \bigcap_{i \in I_{\text{bulk}}} \bigcap_{t \in [0, T]} \{g_i(t) \geq n^{-C_0}\} \cap \left\{ \sup_{|t-s| \leq n^{-A}} |g_i(t) - g_i(s)| \leq n^{-B} \right\}$$

for constants A, B, C_0 chosen as in Lemma 4.1. The Bessel comparison event is $\mathcal{G}_{i,k} := \{\sigma_i(\mathbf{b}, T) > k\mathbf{b}\} \cap \Omega_{\text{RMT}}(T)$, where $\sigma_i(\mathbf{b}, T)$ is the stopping time of Definition 4.8.

Lemma 4.3. *For any $D > 0$, choosing A, B, C_0 large enough, $P(\Omega_{\text{RMT}}(T)^c) \leq n^{-D}$.*

Proof. By Lemma 4.1, the first event in (12) holds with probability $1 - n^{-D-1}$. The modulus-of-continuity and lower bound on g_i follow from Lemma 4.4 and the non-collision property, the latter being a consequence of the log-Sobolev inequality (Theorem 3.2). A union bound over these finitely many events gives the result. $\square$

Lemma 4.4 (Modulus of continuity). *For any $\delta_0, \varepsilon_0 > 0$ and bulk i , $P(\sup_{|t-s| \leq \delta_0} |g_i(t) - g_i(s)| > \varepsilon_0) \leq C \exp(-c\varepsilon_0^2/\delta_0)$.*

Proof. On $\Omega_{\text{RMT}}(T)$, the gap SDE (8) has diffusion coefficient $2/\sqrt{\beta n}$ and total drift bounded by $C(\beta)$ uniformly (this follows directly from the far-field bound $|R_i^{\text{far}}| \leq C/L = O(1)$ (Lemma 4.6) and the near-field bound $|R_i^{\text{near}}| \leq C$ from $\Omega_{\text{RMT}}(T)$ itself, independently of Corollary 4.7). Writing $g_i(t) = g_i(0) + \int_0^t \mu_s ds + (2/\sqrt{\beta n}) W_t$ with $|\mu_s| \leq C$, we have $|g_i(t) - g_i(s)| \leq C|t-s| + (2/\sqrt{\beta n})|W_t - W_s|$. The Brownian increment is sub-Gaussian with variance $|t-s|$, so for $|t-s| \leq \delta_0$: $P(|W_t - W_s| > \varepsilon_0 \sqrt{n}/4) \leq 2 \exp(-\varepsilon_0^2 n / (32\delta_0))$. The drift term contributes $C\delta_0$, absorbable into $\varepsilon_0/4$ for $\delta_0 \leq \varepsilon_0/(4C)$. Choosing $\varepsilon_0 = \varepsilon_0$ and combining gives $P(|g_i(t) - g_i(s)| > \varepsilon_0) \leq C \exp(-c\varepsilon_0^2/\delta_0)$. $\square$

4.3. Far-field drift cancellation.

Lemma 4.5 (Hilbert transform of the semicircle). *For $|x| < 2$,*

$$\text{P.V.} \int_{-2}^2 \frac{\varrho_{\text{sc}}(y)}{x-y} dy = -\frac{x}{2}.$$

This identity reflects the equilibrium condition: the Coulomb force from the semicircle measure at a bulk point x exactly balances the harmonic restoring force.

Proof. The Stieltjes transform of ϱ_{sc} is $m_{\text{sc}}(z) = (-z + \sqrt{z^2 - 4})/2$ for $z \notin [-2, 2]$ (principal branch, $\sqrt{z^2 - 4} \rightarrow z$ as $z \rightarrow \infty$). By the Sokhotski–Plemelj formula, $\text{P.V.} \int \varrho_{\text{sc}}(y)/(x-y) dy = \text{Re}(m_{\text{sc}}(x + i0^+))$ for $|x| < 2$. Since $x^2 - 4 < 0$ for $|x| < 2$, $\sqrt{(x + i0^+)^2 - 4} = i\sqrt{4 - x^2}$, so $m_{\text{sc}}(x + i0^+) = (-x + i\sqrt{4 - x^2})/2$ and $\text{Re}(m_{\text{sc}}(x + i0^+)) = -x/2$. $\square$

Lemma 4.6 (Uniform bound for the far-field drift). *For a bulk index i and locality scale $L = (\log n)^{\alpha_{\text{rig}}+1}$, define the far-field drift (the $|j-i| > L$ part of the third term of (8))*

$$R_i^{\text{far}}(t) := -\frac{1}{n} \sum_{|j-i| > L} \frac{g_i(t)}{(\lambda_{i+1}(t) - \lambda_j(t))(\lambda_i(t) - \lambda_j(t))}.$$

On $\Omega_{\text{RMT}}(T)$, $|R_i^{\text{far}}(t)| \leq C/L$ uniformly in bulk i , $t \leq T$, and n .

Proof. Using the identity $g_i/[(\lambda_{i+1} - \lambda_j)(\lambda_i - \lambda_j)] = 1/(\lambda_i - \lambda_j) - 1/(\lambda_{i+1} - \lambda_j)$:

$$R_i^{\text{far}} = -\frac{1}{n} \sum_{|j-i|>L} \left(\frac{1}{\lambda_i - \lambda_j} - \frac{1}{\lambda_{i+1} - \lambda_j} \right).$$

Write $\lambda_i = \gamma_i + \delta_i$, $|\delta_i| \leq \nu := C(\log n)^{\alpha_{\text{rig}}}/\sqrt{n}$.

Step 1: Leading term at classical locations. At the classical locations γ_j , with $|j-i| = m$ one has $\gamma_{i+m} - \gamma_i \approx m/(n\varrho_{\text{sc}}(\gamma_i))$ (quantile spacing). The contribution of the m -th term to $|S| := |(1/n) \sum_{|j-i|>L} [(\gamma_i - \gamma_j)^{-1} - (\gamma_{i+1} - \gamma_j)^{-1}]|$ is:

$$\frac{1}{n} \cdot \frac{g_i^{\text{cl}}}{|\gamma_{i+m} - \gamma_i| \cdot |\gamma_{i+m} - \gamma_{i+1}|} \asymp \frac{\varrho_{\text{sc}}^2}{m^2}.$$

Summing over $m > L$: $|S| \leq C\varrho_{\text{sc}}^2 \sum_{m>L} m^{-2} \leq C\varrho_{\text{sc}}^2/L = O(1/L)$.

Step 2: First-order corrections in $|\delta_j| \leq \nu$. The Taylor remainder from δ_j contributes $(1/n) \cdot |\delta_j|/|\gamma_j - \gamma_i|^2$ per term. Summing: $(1/n) \sum_{m>L} \nu/(m/(n\varrho_{\text{sc}}))^2 = n\varrho_{\text{sc}}^2 \nu \sum_{m>L} m^{-2} = O(n\varrho_{\text{sc}}^2 \nu/L) = O((\log n)^{-1}) = O(1/L)$.

Step 3. Higher-order Taylor terms contribute $O(1/L^k)$ for $k \geq 2$. Combining: $|R_i^{\text{far}}| \leq C/L = O(1)$. $\square$

Corollary 4.7 (Uniform drift bound). *On $\Omega_{\text{RMT}}(T)$, the total additive drift in the gap SDE is bounded by $C(\beta)$ uniformly in n , bulk i , and $t \leq T$.*

4.4. Stopping time for local control.

Definition 4.8 (Local-control stopping time). For a bulk index i ,

$$(13) \quad \sigma_i(\mathbf{b}, T) := \inf\{t \in [0, T] : \min\{g_{i-1}(t), g_i(t), g_{i+1}(t)\} < c_1 n^{-1} \text{ or } \max\{g_{i-1}(t), g_{i+1}(t)\} > K_0 n^{-1}\},$$

with $g_0 = g_n = +\infty$, c_1 small and K_0 large (both depending only on β). The threshold should be $\gg \varepsilon$ in the near-collision regime that §5 establishes, so that g_i stays well above ε during the Bessel approximation interval. On $\Omega_{\text{RMT}}(T)$ one has $\sigma_i = T$ with probability $1 - O(n^{-D})$.

Note: the thresholds here use the $O(n^{-1})$ typical gap scale (§3.2), consistent with the ε -range derived in §5.

On the event $\mathcal{G}_{i,k} = \{\sigma_i(\mathbf{b}, T) > kb\} \cap \Omega_{\text{RMT}}(T)$, the gap SDE simplifies enough to be coupled to a Bessel process via Girsanov's theorem — the heart of the intended proof, discussed next.

5. THE MARGINAL HITTING-TIME THEOREM, AND A JOINT EXTENSION

This section derives the paper's main result: the exponential asymptotic for a single bulk gap's survival probability, Theorem 5.3 below, together with its general- β form. The derivation hinges on a single dimensional-analysis point worth stating up front, since it fixes every constant that follows: at equilibrium, n eigenvalues spread over the $O(1)$ -support semicircle force nearest-neighbour gaps of size $g_i \asymp 1/n$ (not $n^{-1/2}$, as elementary counting against Theorem 3.1 and the local-semicircle scale of Theorem 3.3 already show, and as §3.2's rescaling $G_i = ng_i$ was designed to make precise). We work throughout with the correctly normalised gap-universality statement $ng_i \rightarrow \text{Gaudin}$ (Theorem 3.4). After establishing the marginal theorem (Steps 1–2), §5.3 (Step 3) turns to the natural strengthening — a joint statement across all n gaps, of which the cover-time Gumbel law would be a corollary — and identifies exactly where it currently resists proof.

5.1. Step 1: the marginal rate formula. By Theorem 3.4 ($ng_i \rightarrow \text{Gaudin}$), the one-block hit probability is

$$(14) \quad q = P(g_i(0) \leq \varepsilon) = P(ng_i \leq n\varepsilon) \sim \kappa_G(n\varepsilon)^3, \quad n\varepsilon \rightarrow 0,$$

the cubic small- s behaviour of the Gaudin CDF evaluated at $s = n\varepsilon$. In particular *rarity* ($q \rightarrow 0$) requires $\varepsilon = o(n^{-1})$.

The excursion-duration and rate computation. The mean excursion duration μ_A below level $E := n\varepsilon$ is, by Brownian scaling of $\text{BES}(\beta + 1)$, a *local* quantity near the threshold E — governed by the excursion’s own local time scale E^2 , not by the typical size of G_i away from E . The block length $\mathbf{b}_{\text{loc}} = n^2\varepsilon^2$ (local time) satisfies $\mathbf{b}_{\text{loc}}/E^2 = 1$ exactly; Kac’s formula then gives $r = q/\mu_A \asymp \kappa' n^2\varepsilon$ for a universal constant $\kappa' = \kappa_G \int_0^\infty h(v)p_G(v)dv$ (the hitting function h and the Gaudin density p_G are scale-independent objects).

An auxiliary estimate: hitting the threshold from a typical starting value. The computation of q above implicitly needs a bound on $P(G_i \text{ hits } E \mid G_i(0) = G_0)$ for $G_0 = O(1)$ the typical value of G_i away from the threshold, on the local-time scale $\delta = \mathbf{b}_{\text{loc}} = n^2\varepsilon^2 \rightarrow 0$ of one block. This is a *small*-time regime for the Bessel process ($\delta/G_0^2 \asymp n^2\varepsilon^2 \rightarrow 0$ whenever $\varepsilon = o(n^{-1})$), governed by a Gaussian tail rather than the large-time (“ever hits”) power-law asymptotic of Proposition 2.3(ii) (that formula applies only when the elapsed time is $\gg G_0^2$, the opposite regime).

Concretely: hitting E from $G_0 = O(1)$ requires first hitting the intermediate level $G_0/2$ (by continuity), and on $[G_0/2, G_0]$ the $\text{BES}(3)$ drift $2/X$ is bounded above by $4/G_0 = O(1)$. Coupling, via the same driving Brownian motion, to $Y_s = G_0 + (4/G_0)s + \sqrt{2}V_s$ — constant drift equal to the maximum of $2/X$ on this range — gives $X_s \leq Y_s$ pathwise for as long as $X \geq G_0/2$, hence $P(X \text{ hits } G_0/2 \text{ by } \delta) \leq P(Y \text{ hits } G_0/2 \text{ by } \delta)$. The right-hand side is given exactly by the classical drifted-Brownian-motion running-minimum formula, and as $\delta \rightarrow 0$ (which holds since $\delta = \mathbf{b}_{\text{loc}} \rightarrow 0$) this is $\asymp \exp(-G_0^2/(16\delta)) = \exp(-c(G_0)/(n^2\varepsilon^2))$ for a constant $c(G_0) = G_0^2/16 > 0$. Since $n^2\varepsilon^2 \rightarrow 0$ throughout the rarity regime, this is *stretched-exponentially* small in n — smaller than any power of n , in particular $o(q) = o((n\varepsilon)^3)$ automatically. So the “hitting from above” contribution is negligible, and (14) gives the one-block probability to the stated order.

The ε -range’s upper boundary is determined by the covariance-decay condition $L/n \ll q^2$ alone (the coupling-error condition, $\varepsilon \ll q$, requiring $C < 3/2$, is strictly weaker and automatically satisfied): with $L = (\log n)^{\alpha_{\text{rig}}+1}$ and $q^2 \asymp n^6\varepsilon^6$, the condition $(\log n)^{\alpha_{\text{rig}}+1}/n \ll n^6\varepsilon^6$ gives $\varepsilon \gg (\log n)^{(\alpha_{\text{rig}}+1)/6}/n^{7/6}$, i.e. $C < 7/6$. Combined with rarity ($C > 1$), the admissible ε -range is

$$(15) \quad n^{-C} \leq \varepsilon = o(n^{-1}), \quad C \in (1, 7/6).$$

Remark 5.1 (Worked numerical example, and a sensitivity check). Take $n = 1000$, $C = 1.1 \in (1, 7/6 \approx 1.167)$, so $\varepsilon = n^{-1.1} \approx 5.01 \times 10^{-4}$ and $n\varepsilon \approx 0.501$. Then:

- $q/\kappa_G = (n\varepsilon)^3 \approx 0.126$ (about one in eight blocks near-collides). For contrast, using the dimensionally incorrect normalisation $\sqrt{n}\varepsilon$ in place of $n\varepsilon$ at this same (n, ε) gives $(\sqrt{1000} \cdot 5.01 \times 10^{-4})^3 \approx 6.3 \times 10^{-4}$, roughly 200× smaller — illustrating how sensitive the theory is to getting the normalisation right, consistent with gaps genuinely being $O(1/n)$ rather than $O(n^{-1/2})$;
- $r/\kappa' = n^2\varepsilon \approx 501$, using the numerical estimate $\kappa' \approx 1.847 \kappa_G$ (this constant is scale-independent, as argued above), $r \approx 926 \kappa_G$;
- purely for illustration, *if* the marginal rate above extended to a joint cover-time law of the conjectural form in §1.1 (an extension not established in this note — see §5.3), one would get $\tau_{\text{cov}}^\varepsilon \approx (\log n)/r \approx 6.908/(926 \kappa_G) \approx 0.00746/\kappa_G$, with Gumbel fluctuation scale $1/r \approx 0.00108/\kappa_G$;

- for contrast, using the incorrect $\sqrt{n}\varepsilon$ normalisation throughout would give $r_{\text{old}}/\kappa' = \sqrt{n}\varepsilon \approx 0.0158$ at this same (n, ε) — smaller by a factor of $\approx 3.2 \times 10^4$ at $n = 1000$, a gap that widens polynomially with n (the two formulas scale as $n^2\varepsilon$ vs. $\sqrt{n}\varepsilon$, ratio $n^{3/2}$). The two normalisations are not close variants of each other; they describe quantitatively very different regimes.

Remark 5.2 (General β). Step 1's argument did not use $\beta = 2$ specifically except through the cubic exponent; redoing it for general $\beta > 1$ (recall $\beta = 1$ is excluded, on grounds unrelated to the normalisation above) gives, from (H_β) evaluated at $s = n\varepsilon$:

$$q_\beta \sim \kappa_\beta(n\varepsilon)^{\beta+1}, \quad r_\beta \asymp \kappa'_\beta n^\beta \varepsilon^{\beta-1}$$

(the block length $\mathbf{b}_{\text{loc}} \asymp n^2\varepsilon^2$ is unchanged for every β , since it is fixed by the threshold scale $E = n\varepsilon$ and the $\text{BES}(\beta+1)$ diffusion coefficient $2/\sqrt{\beta}$, not by β 's effect on G_i 's typical size). The rarity condition is $\varepsilon = o(n^{-1})$ for every β (the exponent $\beta+1$ in q_β does not affect whether $n\varepsilon \rightarrow 0$). Redoing the covariance-decay condition $L/n \ll q_\beta^2$: $q_\beta^2 \asymp n^{2\beta+2}\varepsilon^{2\beta+2}$, so $\varepsilon \gg (\log n)^{(\alpha_{\text{rig}}+1)/(2\beta+2)}/n^{(2\beta+3)/(2\beta+2)}$, i.e.

$$(16) \quad C < C^*(\beta) := \frac{2\beta+3}{2\beta+2} \quad (C^*(2) = \frac{7}{6}, C^*(4) = \frac{11}{10}).$$

The coupling-error condition ($\varepsilon \ll q_\beta$) requires $C < (\beta+1)/\beta$, which satisfies $C^*(\beta) = 1 + \frac{1}{2\beta+2} < 1 + \frac{1}{\beta} = \frac{\beta+1}{\beta}$ for every $\beta > 0$ (since $2\beta+2 > \beta$), so it is never binding — the same pattern as $\beta = 2$. The admissible range is therefore $\varepsilon = n^{-C}$, $C \in (1, C^*(\beta))$, *narrowing* as β grows ($C^*(\beta) \rightarrow 1$ as $\beta \rightarrow \infty$): stronger level repulsion leaves less room in the exponent, consistent with the qualitative picture that larger β makes the theory more delicate. The "hitting from above" resolution (Step 1 above) also generalizes: the comparison argument only used that the $\text{BES}(\beta+1)$ drift $\beta/(2x)$ is $O(1)$ on $[G_0/2, G_0]$ for fixed $G_0 = O(1)$, which holds for every β .

Worked example at $\beta = 4$ (GSE). The $n = 1000$, $C = 1.1$ example used above for $\beta = 2$ is *not* admissible at $\beta = 4$, since $C^*(4) = 11/10 = 1.1$ exactly — this illustrates concretely how the admissible range shrinks with β (at $\beta = 2$, $C = 1.1$ was safely inside $(1, 7/6 \approx 1.167)$; at $\beta = 4$ the same C sits right on the boundary). Taking instead $C = 1.05 < C^*(4)$: $\varepsilon = n^{-1.05} \approx 7.08 \times 10^{-4}$, $n\varepsilon \approx 0.708$, and

$$q_4/\kappa_4 = (n\varepsilon)^5 \approx 0.178, \quad r_4/\kappa'_4 = n^4\varepsilon^3 \approx 355.$$

Both are smaller than the analogous $\beta = 2$ quantities at comparable (n, ε) , quantifying the qualitative statement that stronger repulsion means slower, more threshold-sensitive covering — and the narrower admissible range itself is a new, quantitative signature of that same effect.

Combining (14) with the Girsanov–Bessel coupling and the no-clustering estimate established in Step 2 immediately below (which supply the hypotheses of the Abadi–Galves rare-event theorem) yields the paper's main result.

Theorem 5.3 (Marginal hitting-time asymptotics). *Fix $\beta > 1$ and a bulk index $i \in I_{\text{bulk}}$. Let $K := T/\mathbf{b}_{\text{loc}}$ denote the number of local-time blocks up to time T , where $\mathbf{b}_{\text{loc}} = n^2\varepsilon^2$. For $\varepsilon = n^{-C}$ with $C \in (1, C^*(\beta))$, $C^*(\beta) = (2\beta+3)/(2\beta+2)$ as in (16), the probability that gap i never near-collides before time T satisfies*

$$P(g_i(t) > \varepsilon \ \forall t \in [0, T]) = \exp(-(1+o(1))Kq_\beta) + o(1), \quad q_\beta \sim \kappa_\beta(n\varepsilon)^{\beta+1} \ (n \rightarrow \infty),$$

with κ_β the constant of hypothesis (H_β) . At $\beta = 2$ (the Gaussian case) this recovers $q \sim \kappa_G(n\varepsilon)^3$ as in (14).

The proof occupies the remainder of Step 1 (the rate q_β , already derived above, and the auxiliary hitting-from-above estimate, shown negligible above) and all of Step 2 (the Girsanov TV bound and no-clustering, below). §5.4 gives a second instance of the same theorem for a non-semicircle bulk density (Wishart–OU / Marchenko–Pastur), establishing that the argument is not particular to the Gaussian case.

5.2. Step 2: the Girsanov–Bessel coupling and no-clustering. The general method of §§4 (uniform rigidity, far-field cancellation) is retained above as-is and does not need to change. Checking the two remaining pieces in detail:

Girsanov TV bound. The change of measure removing gap i 's local drift over one block satisfies $\|dQ/dP - 1\|_{L^1} = O(\varepsilon)$ via $\mathbb{E}[(dQ/dP)^2] = \exp(\beta \int_0^{\mathbf{b}_{\text{loc}}} \tilde{R}_i^2 ds) \leq \exp(\beta C_R^2 \varepsilon^2)$, using only $|\tilde{R}_i| \leq C_R/n$ (a rigidity/far-field bound, unrelated to G_i 's typical size) and $\mathbf{b}_{\text{loc}} = n^2 \varepsilon^2$ (fixed by the excursion scale E^2 , a purely local quantity). This computation never references G_i 's global scale, so the per-block TV bound is $O(\varepsilon)$.

No-clustering. Two ingredients are needed: (i) Bessel repulsion after a near-hit, $P(G_i(s_0 + \mathbf{b}_{\text{loc}}) \geq c_\beta E \mid G_i(s_0) = E)$ — a statement purely about ratios relative to E (Brownian self-similarity of $\text{BES}(\beta+1)$), unaffected by G_i 's global scale; and (ii) hitting E again from $c_\beta E$ within one more block, using Proposition 2.3(ii) with $x_0 = c_\beta E$ and $\delta = \mathbf{b}_{\text{loc}} = E^2$, so $\delta/x_0^2 = 1/c_\beta^2 = O(1)$ — the intermediate regime the hitting formula is designed for, giving $p'_c = c_\beta^{-(\beta-1)} < 1$. Neither step involves G_i 's typical value G_0 at all (only the local, E -relative behaviour after a near-hit).

Both pieces of Step 2 hold with the same constants at every C in the admissible range; only Step 1's formulas for q_β and the ε -range feed into them. This completes the proof of Theorem 5.3.

Why this does not extend directly to a joint statement. The per-gap coupling bound $\delta_n = KC\varepsilon \asymp (\log n)n^{2C-3} \rightarrow 0$ for $C \in (1, 7/6)$, but a union bound over all n gaps, $n\delta_n \asymp (\log n)n^{2C-2} \rightarrow \infty$ for $C > 1$, diverges. So the natural approach to a joint statement — couple every gap to an independent Bessel proxy and transfer a Poisson approximation from the proxy system to the true joint count — cannot be salvaged simply by re-deriving constants; §5.3 takes this up directly.

5.3. Step 3: towards a joint theorem, and the cover-time question. A cover-time Gumbel law for the bulk (Aldous–Ding–Lee–Peres-type [3, 6]) is the natural strengthening of Theorem 5.3: it requires not a single gap's marginal survival rate but a joint statement across all n gaps simultaneously, of Poisson-approximation type, from which the maximum over $i \in I_{\text{bulk}}$ of the individual near-collision times could be extracted by the usual extreme-value machinery. We record here exactly how far the natural strategy for such a joint statement can be pushed, and where it currently stops.

Coupling each gap independently to an idealized, exactly-independent Bessel proxy system, then transferring the resulting exact Poisson law back to the real joint count $W(T) = \sum_i I_i(T)$, provably fails. Four independent computations (a simultaneous union bound over all gaps and blocks; a per-gap union bound; a direct $\mathbb{E}|W - W^{\text{Bes}}|$ bound; a $\text{Var}(W - W^{\text{Bes}})$ bound) all reduce to the same divergent quantity, ruling out that specific transfer strategy rather than merely failing to complete it. Two candidate routes remain open:

- A genuinely sharper Girsanov TV bound (smaller than the per-block $C\varepsilon$ that Step 2 will reproduce in some corrected form) might survive the union bound needed for the transfer. Whether such a bound is available was not investigated.
- A direct Chen–Stein local-dependence argument (Arratia–Goldstein–Gordon [2]) applied to the real indicators $\{I_i(T)\}$, bypassing the Bessel proxy entirely. The far-pair contribution (gaps separated by more than the locality scale L) is handled

by covariance decay via mixing (Theorem 3.2, no Bessel coupling needed). The near-pair contribution was initially thought to hit a single obstruction across the whole locality scale L ; on closer inspection the shared-noise problem is narrower than that, though not yet fully resolved:

The noise-sharing problem is exactly localized to $|i - j| = 1$. The gap SDE's noise term for g_i is $\sqrt{2/(\beta n)}(dB_{i+1} - dB_i)$; for g_j it is $\sqrt{2/(\beta n)}(dB_{j+1} - dB_j)$. These involve a common dB_k term iff $\{i, i+1\} \cap \{j, j+1\} \neq \emptyset$, i.e. iff $|i - j| = 1$ — for $|i - j| \geq 2$ the driving noises are *exactly independent* (disjoint Brownian increments), with no shared-noise obstruction at all. So the genuinely new difficulty is confined to immediate neighbours, not a cluster of size $O(L)$.

For $|i - j| = 1$, a direct computation gives $\text{Cov}(dg_i, dg_{i+1}) = -\frac{1}{2}\sqrt{\text{Var}(dg_i)\text{Var}(dg_{i+1})}$ (correlation exactly $-1/2$, independent of n, β, i): writing out $\text{Cov}(dB_{i+1} - dB_i, dB_{i+2} - dB_{i+1}) = -dt$ against $\text{Var}(dB_{i+1} - dB_i) = 2dt$ gives the ratio $-1/2$ directly. The natural orthogonal transform $U := (g_i + g_{i+1})/\sqrt{2}$, $V := (g_i - g_{i+1})/\sqrt{2}$ **decorrelates the noise exactly**: $\text{Cov}(dU, dV) \propto \text{Cov}(dB_{i+2} - dB_i, 2dB_{i+1} - dB_i - dB_{i+2}) = -dt + dt = 0$ (the two surviving terms, from $\text{Cov}(dB_i, dB_i)$ and $-\text{Cov}(dB_{i+2}, dB_{i+2})$, cancel exactly). So U, V are driven by independent Brownian motions, and the coupled 2-gap problem reduces to a bounded (2-dimensional, not $O(L)$ -dimensional) diffusion problem with decorrelated noise — the same kind of Girsanov/Bessel-comparison machinery used for a single gap should apply here too, in principle.

Translating this into an actual bound on $P(I_i(T) = 1, I_{i+1}(T) = 1)$ requires expressing the joint survival event $\{g_i > \varepsilon\} \cap \{g_{i+1} > \varepsilon\}$ in terms of (U, V) — it becomes $\{U > \varepsilon\sqrt{2}\} \cap \{|V| < U - \varepsilon\sqrt{2}\}$, a wedge, not a product region. The noise being decorrelated is not by itself enough: on the rescaled gaps $G_i := ng_i$, the drift terms $2/G_i, 2/G_{i+1}$ transform to

$$\text{drift}(U) = \frac{4U}{U^2 - V^2}, \quad \text{drift}(V) = \frac{-4V}{U^2 - V^2} \quad (\text{verified directly by substitution}),$$

which couples U and V through the drift even though the noise is independent — so (U, V) is not simply a product of two 1-dimensional diffusions.

A further, partial result: consider instead the product $Z := G_i G_{i+1} = (U^2 - V^2)/2$. By Itô's product rule, using $\text{Cov}(dG_i, dG_{i+1}) = -\frac{1}{2}\sqrt{\text{Var}(dG_i)\text{Var}(dG_{i+1})} = -\frac{2}{\beta}ds$ (the same $-1/2$ correlation, carried through the rescaling, with $\text{Var}(dG_i) = \frac{4}{\beta}ds$ each),

$$dZ = \left[\frac{2(G_i^2 + G_{i+1}^2)}{Z} - \frac{2}{\beta} \right] ds + (\text{noise}), \quad \text{Var}(\text{noise}) = \frac{4}{\beta} [G_i^2 + G_{i+1}^2 - Z] ds$$

(the drift term verified directly by finite-difference substitution). By AM-GM, $G_i^2 + G_{i+1}^2 \geq 2Z$, so

$$(17) \quad \text{drift}(Z) \geq \frac{2 \cdot 2Z}{Z} - \frac{2}{\beta} = 4 - \frac{2}{\beta} =: c_\beta > 0 \quad \text{for all } \beta > \frac{1}{2},$$

regardless of Z 's value or the ratio G_i/G_{i+1} : the product of two adjacent gaps has a systematic, uniformly bounded-below positive drift (at $\beta = 2, c_\beta = 3$). This is a genuine new structural fact, not present in the single-gap theory, suggesting Z hitting small values (both gaps simultaneously small) is suppressed by an extra, persistent push — analogous in spirit to the single-gap Bessel repulsion.

A complete bound, for $\beta \geq 2$, via a supermartingale. Z 's drift and diffusion both depend on $S := G_i^2 + G_{i+1}^2$, not on Z alone, which is what makes a direct comparison-theorem argument for Z awkward. Rather than control S

separately (e.g. by sandwiching it via its own, easily-checked constant-drift SDE $dS = (8 + 8/\beta)ds + (\text{noise})$, $\text{Var}(\text{noise}) = \frac{16}{\beta}(S - Z)ds$, and then comparing Z 's drift on the resulting high-probability event — an approach that can be made to work but is cumbersome), it is cleaner to analyse the reciprocal $1/Z$ directly, without approximating S at all. By Itô's formula applied to $f(Z) = 1/Z$, using the exact drift and diffusion of Z recorded above,

$$(18) \quad d(1/Z) = \left[\frac{S(4 - 2\beta)}{\beta Z^3} - \frac{2}{\beta Z^2} \right] ds + (\text{martingale part}),$$

(verified directly by finite-difference substitution at several (β, G_i, G_{i+1}) points, matching the closed form to within Monte Carlo error). For $\beta \geq 2$ the coefficient $4 - 2\beta \leq 0$, so — since $S \geq 0$ always — the bracketed drift is $\leq -2/(\beta Z^2) < 0$ *unconditionally*, with no need to restrict to any high-probability event for S : $1/Z_s$ is a nonnegative local supermartingale for every s , hence (a nonnegative local martingale is automatically a true supermartingale, by Fatou's lemma along the localizing sequence $\tau_m := \inf\{s : Z_s \leq 1/m\}$, using that $Z_s = G_i(s)G_{i+1}(s) > 0$ a.s. for every s since each factor is a.s. positive for $\beta > 1$) a genuine supermartingale. Ville's maximal inequality for nonnegative supermartingales then gives, for *every* $\zeta > 0$ and *every* time horizon $T > 0$ (not just one block),

$$(19) \quad P\left(\inf_{0 \leq s \leq T} Z_s \leq \zeta\right) = P\left(\sup_{0 \leq s \leq T} \frac{1}{Z_s} \geq \frac{1}{\zeta}\right) \leq \frac{\zeta}{Z_0}.$$

Since $\{g_i(s) \leq \varepsilon\} \cap \{g_{i+1}(s) \leq \varepsilon\}$ at any single time s forces $Z(s) = G_i(s)G_{i+1}(s) \leq E^2$ (with $E = n\varepsilon$), (19) with $\zeta = E^2$ gives an explicit, unconditional bound on the near-pair joint near-collision probability,

$$(20) \quad P(\exists s \leq T : g_i(s) \leq \varepsilon \text{ and } g_{i+1}(s) \leq \varepsilon) \leq \frac{(n\varepsilon)^2}{Z_0}, \quad \beta \geq 2,$$

with $Z_0 = G_i(0)G_{i+1}(0) = O(1)$ the (typical, $O(1)$) starting value. This is a complete, closed-form resolution of the target left as "not completed" in an earlier version of this argument (there, S was frozen at S_0 as a heuristic simplification, which for $1 < \beta < 2$ suggested — incorrectly, since the freezing itself was never justified — an exactly critical effective dimension at $\beta = 2$; (19) settles the question directly and rigorously for $\beta \geq 2$, with no freezing anywhere in the argument, superseding that heuristic). Both (18) and the resulting tail bound (19) were checked by direct simulation of the true coupled (G_i, G_{i+1}) pair at $\beta = 2$ (20,000 trajectories, $Z_0 = 2.25$, time horizons up to $T = 20$): the empirical probability of Z ever falling below a threshold ζ stayed below the bound ζ/Z_0 at every $\zeta \in \{0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.4\}$ tested, and came within 1% of it at the smallest ζ — consistent with $1/Z$ behaving, near the origin, like the classical inverse-BES(3) process (for which a/x_0 is not just an upper bound but the exact hitting probability), for which (19) is known to be asymptotically sharp.

What this does, and does not, complete. For $1 < \beta < 2$ the coefficient $4 - 2\beta$ in (18) is positive, so the argument does not directly give a supermartingale, and that range remains open. For $\beta \geq 2$ (in particular $\beta = 2$, the case of primary interest), (20) is a genuine, unconditional, verified bound — but it is a global (Ville/Doob-type) estimate over the whole time horizon T , not a small-time, large-deviation one, and by that measure it is unlikely to be sharp enough on its own to close the full joint theorem: at order ε^2 , it is polynomially *larger* than the product $q^2 \sim \kappa_G^2(n\varepsilon)^6$ of individual failure probabilities that an independence-type local-dependence bound would ultimately need to match. A

further refinement restricted to a single local-time block $\delta = \mathbf{b}_{\text{loc}} \rightarrow 0$ (exploiting the short time horizon, in the same spirit as the small-time argument that resolved the "hitting from above" correction in Step 1) would plausibly sharpen (20) substantially, but was not carried out here. The overlapping-window issue for $2 \leq |i - j| \leq L$ (drift/Coulomb coupling through the full eigenvalue configuration, unrelated to noise-sharing) remains a separate, untouched question.

5.4. An application: a single risk-factor pair. *Cover time* and *minimum* statements (the last, resp. first, gap to near-collide) both need control over *all* n gaps jointly, which is exactly the open step of §5.3. A statement about a single, *fixed* gap needs only Theorem 5.3's marginal survival-rate result — established for every $\beta > 1$ per Remark 5.2 — with no joint approximation at all; the application below is a second, independent instance of that theorem.

Setup. Let $\hat{\Sigma}_t = X_t X_t^\top / k$ be a sample covariance matrix under Wishart-Ornstein-Uhlenbeck dynamics ($X_t \in \mathbb{R}^{n \times k}$, $dX_t = -\frac{1}{2}X_t dt + n^{-1/2}dW_t$), $n/k \rightarrow \gamma \in (0, 1)$, bulk eigenvalue density $\varrho_{\text{MP}}(\cdot; \gamma)$ (Marchenko–Pastur). *The same elementary counting argument underlying Theorem 3.4* applies here without modification: n eigenvalues over an $O(1)$ -support density force local spacing $g_i \asymp 1/(n\varrho_{\text{MP}}(x_0))$, not $O(n^{-1/2})$ — this is not specific to the semicircle law. At the bulk centre $x_0 = \sigma^2$ (population variance), a direct computation gives

$$(21) \quad \varrho_{\text{MP}}(\sigma^2; \gamma) = \frac{1}{2\pi\sigma^2} \sqrt{\frac{4 - \gamma}{\gamma}}$$

(verified numerically against the defining formula for $\gamma = 0.3, 0.5, 0.7$; e.g. $\varrho_{\text{MP}}(1; 0.5) \approx 0.421$).

The safe, marginal statement. For a *fixed* bulk pair (σ_i, σ_{i+1}) (two specific "risk factors"), the same Abadi–Galves/Kac's-formula argument used for the marginal single-gap rate gives

$$P(\sigma_{i+1}(t) - \sigma_i(t) > \varepsilon \ \forall t \leq T) = \exp(-(1 + o(1))Kq_i) + o(1), \quad q_i \sim \kappa_G(n\varrho_{\text{MP}}(x_0)\varepsilon)^3,$$

by the same Kac's-formula argument as Step 1, with the semicircle density replaced by $\varrho_{\text{MP}}(x_0; \gamma)$ at the bulk point x_0 of interest. This is a statement about *one specific pair of eigenvalues*, not the worst case over all $\binom{n}{1}$ bulk pairs — it needs no control of joint dependence across different i , so Step 3's obstruction is simply not engaged.

What is not verified here. The precise constant in (H_β) -type statements for non-semicircle bulk densities is standardly stated with an explicit local-density factor (the "unfolded" gap $n\varrho(x_0)g_i$, not just ng_i); whether this factor is correctly threaded through κ_G above, or partially absorbed into it, was not checked with the same care as the semicircle case — this is a specific, identified gap, not a vague one, and should be resolved before quoting an exact numerical constant.

A worked example, and a sensitivity check. Take $n = 500$, $\gamma = 0.5$, $\varepsilon = 10^{-3}$, so $n\varrho_{\text{MP}}(\sigma^2; 0.5)\varepsilon \approx 0.211$. For contrast, using the (dimensionally incorrect) normalisation $\sqrt{n}\varepsilon$ in place of $n\varrho_{\text{MP}}(\sigma^2; 0.5)\varepsilon$ gives ≈ 0.0224 at this same (n, γ, ε) — a factor of ≈ 9.4 , growing polynomially with n exactly as in Remark 5.1. The two scalings are not close variants of one another, which is why getting the normalisation right (the point of §3.2) matters quantitatively and not just qualitatively for an application such as this one: near-collision of a specific risk-factor pair is far more likely, at fixed tolerance and horizon, than the naive scaling would suggest. The exact quantitative restatement needs the local-density point above resolved first, and is not attempted here.

5.5. Ingredients shared by the marginal theorem and the open joint extension. Both Theorem 5.3 and any future joint statement rest on the same underlying machinery:

the Dyson-OU SDE and its equilibrium measure (§3); the microscopic rescaling $G_i = ng_i$ (§3.2); the external rigidity, log-Sobolev, and local-semicircle inputs (Theorems 3.1–3.3); the general theory of Bessel processes and level repulsion (§2); and the high-probability control machinery (§4). None of this needs to be revisited in extending the theory to a joint statement — the open step is specifically the one identified in §5.3.

5.6. Summary. Theorem 5.3 (Steps 1–2) establishes the marginal hitting-time asymptotic for every $\beta > 1$, with an explicit, checked constant and ε -range; §5.4 gives a second, independent instance for a non-semicircle bulk density. The natural strengthening to a joint statement across all n gaps — of which a cover-time Gumbel law would be a corollary — reduces, after §5.3, to the near-pair contribution at unit spatial separation. For this, a uniformly positive drift bound on the product $Z = G_i G_{i+1}$ (17) together with an exact supermartingale property of $1/Z$ for $\beta \geq 2$ yields a complete, closed-form, numerically verified bound (20) on the near-pair joint near-collision probability — resolving, for $\beta \geq 2$, the specific comparison-theorem target left open in earlier work on this note. That bound is a global (Ville-type) estimate rather than a small-time one, and is likely not yet sharp enough by itself to close the full joint theorem (§5.3 quantifies the remaining gap in orders of ε); $1 < \beta < 2$ is untouched by this argument and remains open.

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